



**Laboratorium
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Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-MultiVendor

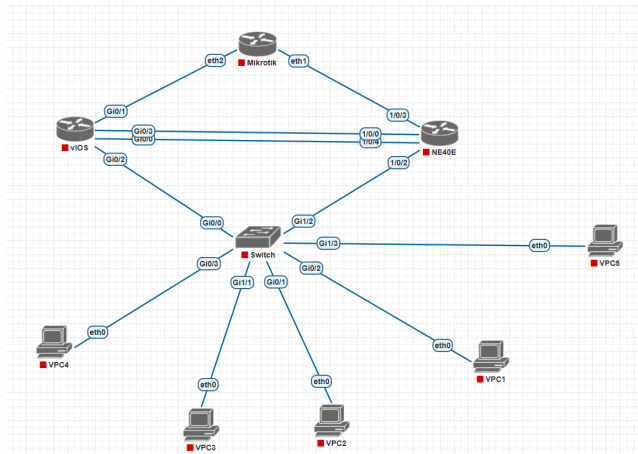
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2026

1 Langkah-Langkah Percobaan

1.1 Konfigrasi Cisco Switch

1. Menyusun topologi sesuai dengan modul 5.



Gambar 1: topologi

2. Mengonfigurasi Cisco Switch untuk membuat VLAN 10, 20, 30, 40 dan memastikan statusnya aktif.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	Finance	active	
20	HR	active	
30	IT	active	
40	Marketing	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Gambar 2: show vlan brief

3. Mengonfigurasi access port untuk setiap vlan sesuai dengan interface pada cisco switch.
4. Mengonfigurasi trunk port dari switch menuju cisco router untuk vlan 10, 20, 30, dan 40.

```
Switch#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/0	Vlans allowed on trunk			
	10,20,30,40			
Gi0/0	Vlans allowed and active in management domain			
	10,20,30,40			
Gi0/0	Vlans in spanning tree forwarding state and not pruned			
	none			

Gambar 3: show interfaces trunk

5. Menyimpan semua konfigurasi.

```
Switch#write memory
Building configuration...
Compressed configuration from 3364 bytes to 1652 bytes[OK]
Switch#
*Jun 5 08:38:52.282: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun 5 08:38:52.982: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
```

Gambar 4: write memory

1.2 Konfigurasi Cisco Router

1. Mengonfigurasi interface trunk ke switch dengan interface yang terhubung adalah gig0/2.
2. Mengonfigurasi subinterface VLAN 10, 20, 30, dan 40.
3. Mengonfigurasi DHCP-Server untuk VLAN 10 dan 20.
4. Mengonfigurasi IP Static pada VPCS untuk VLAN 30 dan 40.
5. Mengecek interface.

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down down
GigabitEthernet0/1    unassigned      YES unset  administratively down down
GigabitEthernet0/2    unassigned      YES unset  up      up
GigabitEthernet0/2.10 192.168.10.1    YES manual  up      up
GigabitEthernet0/2.20 192.168.20.1    YES manual  up      up
GigabitEthernet0/2.30 192.168.30.1    YES manual  up      up
GigabitEthernet0/2.40 192.168.40.1    YES manual  up      up
GigabitEthernet0/3    unassigned      YES unset  administratively down down
GigabitEthernet0/4    unassigned      YES unset  administratively down down
```

Gambar 5: show ip interface brief

6. Mengecek DHCP pool.

```
Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1
```

Gambar 6: show ip dhcp pool

7. Mengecek client DHCP yang mendapatkan IP.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.6e Jun 06 2026 08:51 AM    Automatic
192.168.20.11    0100.5079.6668.6d Jun 06 2026 08:51 AM    Automatic
```

Gambar 7: show ip dhcp binding

8. Masuk ke PC VLAN 10 untuk meminta IP secara DHCP dan memastikannya berhasil.

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE : 86392, 86400/43200/75600
MAC        : 00:50:79:66:68:6e
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
```

Gambar 8: show ip

9. Menguji dari PC VLAN 10 ping gateway 192.168.10.1.

```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.208 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.336 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.133 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.763 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.018 ms
```

Gambar 9: ping 192.168.10.1

10. Masuk ke PC VLAN 20 untuk meminta IP secara DHCP dan memastikannya berhasil.

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86398, 86400/43200/75600
MAC        : 00:50:79:66:68:6d
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
```

Gambar 10: show ip

11. Menguji dari PC VLAN 20 ping gateway 192.168.20.1.

```
VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.000 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.293 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.633 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.136 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=2.776 ms
```

Gambar 11: ping 192.168.20.1

12. Menguji dari PC VLAN 30 ping gateway 192.168.30.1.

```
VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.925 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.565 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=3.035 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=5.439 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.268 ms
```

Gambar 12: ping 192.168.30.1

13. Menguji dari PC VLAN 40 ping gateway 192.168.40.1.

```
VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.703 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.404 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.323 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.380 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=3.613 ms
```

Gambar 13: ping 192.168.40.1

14. Menguji ping antar-VLAN dari VLAN 10.

```
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=5.048 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=4.911 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=5.151 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=5.608 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=4.899 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=4.810 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.967 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.851 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.641 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=5.020 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=4.762 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=3.950 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.696 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.910 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=4.102 ms
```

Gambar 14: ping 192.168.20.11, ping 192.168.30.10, dan ping 192.168.40.10

15. Masuk ke cisco router lagi untuk mengecek IP DHCP binding.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.6e  Jun 06 2026 08:53 AM    Automatic
192.168.20.11    0100.5079.6668.6d  Jun 06 2026 08:54 AM    Automatic
```

Gambar 15: show ip dhcp binding

1.3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Mengonfigurasi IP address untuk LAN Huawei.
2. Mengonfigurasi IP VPC LAN Huawei.

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:76
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
```

Gambar 16: show ip

3. Menguji ping gateway huawei.

```
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=9.810 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.739 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=4.161 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.166 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.408 ms
```

Gambar 17: ping 192.168.50.1

4. Mengonfigurasi IP link cisco dengan huawei link 1 dan link 2.

5. Mengonfigurasi IP link huaweui dengan mikrotik.

6. Mengonfigurasi IP link cisco dengan mikrotik.

7. Memverifikasi interface pada cisco router.

```
Router#show ip interface brief

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.11.1     YES manual up          up
GigabitEthernet0/1  10.10.30.1     YES manual up          up
GigabitEthernet0/2  unassigned     YES unset  up          up
GigabitEthernet0/2.10 192.168.10.1  YES manual up          up
GigabitEthernet0/2.20 192.168.20.1  YES manual up          up
GigabitEthernet0/2.30 192.168.30.1  YES manual up          up
GigabitEthernet0/2.40 192.168.40.1  YES manual up          up
GigabitEthernet0/3   10.10.10.1     YES manual up          up
GigabitEthernet0/4   unassigned     YES unset  administratively down down
```

Gambar 18: show ip interface brief

8. Memverifikasi interface pada huawei router.

```
[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface      IP Address/Mask  Physical  Protocol  VPN
Ethernet1/0/0  10.10.10.2/30    up        up        --
Ethernet1/0/0.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/1     unassigned       up        down      --
Ethernet1/0/1.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/2     192.168.50.1/24  up        up        --
Ethernet1/0/2.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/3     10.10.20.1/30    up        up        --
Ethernet1/0/3.4094 128.1.138.137/16 up        up        l3vpn
Ethernet1/0/4     10.10.11.2/30    up        up        --
Ethernet1/0/4.4094 128.1.138.137/16 up        up        l3vpn
---- More ----
```

Gambar 19: display ip interface brief

9. Memverifikasi interface pada mikrotik.

```
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
#   ADDRESS      NETWORK    INTERFACE
0   ;;; TO-HUAWEI 10.10.20.0 ether1
    10.10.20.2/30
1   ;;; TO-CISCO  10.10.30.0 ether2
    10.10.30.2/30
[admin@MikroTik] >
```

Gambar 20: /ip address print

10. Memverifikasi ping antar-router dari cisco router.

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/3 ms
Router#
```

Gambar 21: ping 10.10.10.2, ping 10.10.11.2, dan ping 10.10.30.2

11. Memverifikasi ping antar-router dari router huawei.

```
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms
```

Gambar 22: ping 10.10.10.1, ping 10.10.11.1, dan ping 10.10.20.2

12. Memverifikasi ping antar-router dari router mikrotik.

```

[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                               SIZE TTL TIME  STATUS
0 10.10.20.1                           56 255 3ms
1 10.10.20.1                           56 255 1ms
2 10.10.20.1                           56 255 1ms
3 10.10.20.1                           56 255 1ms
4 10.10.20.1                           56 255 1ms
5 10.10.20.1                           56 255 2ms
6 10.10.20.1                           56 255 1ms
7 10.10.20.1                           56 255 1ms
sent=8 received=8 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=3ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                               SIZE TTL TIME  STATUS
0 10.10.30.1                           56 255 2ms
1 10.10.30.1                           56 255 1ms
2 10.10.30.1                           56 255 4ms
3 10.10.30.1                           56 255 6ms
4 10.10.30.1                           56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=2ms max-rtt=6ms

```

Gambar 23: ping 10.10.20.1 dan ping 10.10.30.1

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengonfigurasi OSPF pada cisco router, dan mengatur OSPF cost-nya pada interface cisco.

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

```

Gambar 24: show ip protocols

2. Mengonfigurasi OSPF pada hauwei router, dan mengatur OSPF cost-nya pada interface huawei.

```

[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.10.1       BDR: 10.10.10.2     MTU: 1500
  Dead timer due in 38 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m08s
  Neighbor Up Time : 2026-06-05 09:27:36
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.11.1       BDR: 10.10.11.2     MTU: 1500
  Dead timer due in 39 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m08s
  Neighbor Up Time : 2026-06-05 09:27:36
  Authentication Sequence: [ 0 ]

```

Gambar 25: display ospf peer

3. Memastikan huawei peer dengan cisco melalui 2 link utama.

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto Pre  Cost   Flags NextHop         Interface
-----
10.10.30.0/30       OSPF 10   110    D  10.10.11.1 Ethernet1/0/4
10.10.30.0/30       OSPF 10   110    D  10.10.10.1 Ethernet1/0/0
192.168.10.0/24      OSPF 10   11     D  10.10.11.1 Ethernet1/0/4
192.168.10.0/24      OSPF 10   11     D  10.10.10.1 Ethernet1/0/0
192.168.20.0/24      OSPF 10   11     D  10.10.11.1 Ethernet1/0/4
192.168.20.0/24      OSPF 10   11     D  10.10.10.1 Ethernet1/0/0
192.168.30.0/24      OSPF 10   11     D  10.10.11.1 Ethernet1/0/4
192.168.30.0/24      OSPF 10   11     D  10.10.10.1 Ethernet1/0/0
192.168.40.0/24      OSPF 10   11     D  10.10.11.1 Ethernet1/0/4
192.168.40.0/24      OSPF 10   11     D  10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto Pre  Cost   Flags NextHop         Interface
-----
10.10.10.0/30        OSPF 10   10     10.10.10.2 Ethernet1/0/0
10.10.11.0/30        OSPF 10   10     10.10.11.2 Ethernet1/0/4
10.10.20.0/30        OSPF 10  100    10.10.20.1 Ethernet1/0/3
192.168.50.0/24      OSPF 10   1     192.168.50.1 Ethernet1/0/2

```

Gambar 26: display ip routing-table protocol ospf

4. Memastikan huawei sudah mendapatkan network dari cisco.

```

[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination      Cost    Type    NextHop    AdvRouter    Area
-----
10.10.10.0/30    10      Direct  10.10.10.2  2.2.2.2      0.0.0.0
10.10.11.0/30    10      Direct  10.10.11.2  2.2.2.2      0.0.0.0
10.10.20.0/30    100     Direct  10.10.20.1  2.2.2.2      0.0.0.0
10.10.30.0/30    110     Stub    10.10.11.1  1.1.1.1      0.0.0.0
10.10.30.0/30    110     Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.10.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.10.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.20.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.20.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.30.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.30.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.40.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.40.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.50.0/24  1       Direct  192.168.50.1 2.2.2.2      0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination      Cost    Type    NextHop    AdvRouter    Area
-----
128.1.138.137/32 0       Direct  128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0

```

Gambar 27: display ospf routing

5. Mengonfigurasi OSPF pada mikrotik, sebagai jalur backup antara cisco dan huawei.

```

[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=13s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=1 ls-requests=0 db-summaries=0 adjacency=9s

```

Gambar 28: /routing ospf neighbor print

6. Mengecek routing table OSPF.

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADo  10.10.10.0/30      10.10.30.1    110
1 ADo  10.10.11.0/30      10.10.20.1    110
2 ADo  192.168.10.0/24      10.10.30.1    110
3 ADo  192.168.20.0/24      10.10.30.1    110
4 ADo  192.168.30.0/24      10.10.30.1    110
5 ADo  192.168.40.0/24      10.10.30.1    110
6 ADo  192.168.50.0/24      10.10.20.1    110
```

Gambar 29: /ip route print where ospf

7. Memverifikasi OSPF neighbor pada cisco router.

```
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:37 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:35 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:32 10.10.10.2 GigabitEthernet0/3
```

Gambar 30: show ip ospf neighbor

8. Memverifikasi OSPF neighbor pada huawei router.

```
[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00h05m41s
Neighbor Up Time : 2026-06-05 09:27:36
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h01m36s
Neighbor Up Time : 2026-06-05 09:31:42
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h05m41s
Neighbor Up Time : 2026-06-05 09:27:36
Authentication Sequence: [ 0 ]
```

Gambar 31: display ospf peer

9. Memverifikasi OSPF neighbor pada mikrotik.

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m13s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m9s
```

Gambar 32: /routing ospf neighbor print

10. Memverifikasi route menuju lan huawei dari cisco router, serta mengecek semua route ospf.

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:06:46 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:06:46 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:06:46 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:02:49, GigabitEthernet0/0
                    [110/110] via 10.10.10.2, 00:02:49, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:07:02, GigabitEthernet0/0
                    [110/11] via 10.10.10.2, 00:07:02, GigabitEthernet0/3
```

Gambar 33: show ip route 192.168.50.0 dan show ip route ospf

11. Mengecek semua route ospf huawei router.

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto Pre Cost      Flags NextHop        Interface
-----
10.10.30.0/30       OSPF  10  110          D  10.10.11.1    Ethernet1/0/4
10.10.10.0/30       OSPF  10  110          D  10.10.10.1    Ethernet1/0/0
192.168.10.0/24     OSPF  10  11           D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24     OSPF  10  11           D  10.10.10.1    Ethernet1/0/0
192.168.30.0/24     OSPF  10  11           D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24     OSPF  10  11           D  10.10.10.1    Ethernet1/0/0
192.168.50.0/24     OSPF  10  11           D  10.10.11.1    Ethernet1/0/4
192.168.50.0/24     OSPF  10  11           D  10.10.10.1    Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto Pre Cost      Flags NextHop        Interface
-----
10.10.10.0/30       OSPF  10  10           10.10.10.2    Ethernet1/0/0
10.10.11.0/30       OSPF  10  10           10.10.11.2    Ethernet1/0/4
10.10.20.0/30       OSPF  10  100          10.10.20.1    Ethernet1/0/3
192.168.50.0/24     OSPF  10  1            192.168.50.1  Ethernet1/0/2
```

Gambar 34: display-ip routing table protocol ospf

12. Memastikan mikrotik memiliki route ospf ke cisco dan huawei.

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto Pre Cost      Flags NextHop        Interface
-----
10.10.30.0/30       OSPF  10  110          D  10.10.11.1    Ethernet1/0/4
10.10.10.0/30       OSPF  10  110          D  10.10.10.1    Ethernet1/0/0
192.168.10.0/24     OSPF  10  11           D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24     OSPF  10  11           D  10.10.10.1    Ethernet1/0/0
192.168.30.0/24     OSPF  10  11           D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24     OSPF  10  11           D  10.10.10.1    Ethernet1/0/0
192.168.50.0/24     OSPF  10  11           D  10.10.11.1    Ethernet1/0/4
192.168.50.0/24     OSPF  10  11           D  10.10.10.1    Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto Pre Cost      Flags NextHop        Interface
-----
10.10.10.0/30       OSPF  10  10           10.10.10.2    Ethernet1/0/0
10.10.11.0/30       OSPF  10  10           10.10.11.2    Ethernet1/0/4
10.10.20.0/30       OSPF  10  100          10.10.20.1    Ethernet1/0/3
192.168.50.0/24     OSPF  10  1            192.168.50.1  Ethernet1/0/2
```

Gambar 35: /ip route print where ospf

13. Menguji tes ping dari PC VLAN 10 menuju LAN huawei.

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=11.543 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.393 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.935 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.005 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.528 ms
```

Gambar 36: ping 192.168.50.10

14. Menguji tes ping dari PC VLAN 20 menuju LAN huawei.

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.700 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.413 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.865 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.170 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.501 ms
```

Gambar 37: ping 192.168.50.10

15. Menguji tes ping dari PC VLAN 30 menuju LAN huawei.

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.571 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.382 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.613 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.063 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.222 ms
```

Gambar 38: ping 192.168.50.10

16. Menguji tes ping dari PC VLAN 40 menuju LAN huawei.

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.021 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.448 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.928 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.385 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.447 ms
```

Gambar 39: ping 192.168.50.10

17. Menguji tes ping dari PC LAN huawei ke IP VLAN cisco.

```

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=10.664 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=6.840 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=5.232 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=6.193 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=5.597 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=14.587 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=6.830 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=5.803 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.150 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=8.657 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=11.039 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=7.986 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.669 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=7.409 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=5.744 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=5.595 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=6.390 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=7.924 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=5.676 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=5.837 ms

```

Gambar 40: ping 192.168.10.11, ping 192.168.20.11, ping 192.168.30.10, dan ping 192.168.40.10

1.5 Pengujian Failover OSPF

1. Melakukan trace dari PC VLAN menuju LAN hauwei.

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.732 ms  5.263 ms  3.289 ms
 2  10.10.10.2     8.361 ms  4.759 ms  4.434 ms
 3  *192.168.50.10 11.030 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 41: trace 192.168.50.10

2. Mematikan salah satu link cisco-huawei, lalu melakukan trace dari PC VLAN menuju LAN hua-wei.

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.326 ms  5.060 ms  3.293 ms
 2  10.10.11.2     6.736 ms  4.479 ms  4.193 ms
 3  *192.168.50.10 5.529 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 42: trace 192.168.50.10

3. Mematikan link cisco-huawei yang kedua, dan melakuakn trace lagi.

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.090 ms  2.903 ms  3.369 ms
 2  10.10.30.2     6.229 ms  5.606 ms  2.770 ms
 3  10.10.20.1     8.770 ms  5.739 ms  6.296 ms
 4  *192.168.50.10 11.768 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 43: trace 192.168.50.10

4. Mengaktifkan kembali dua link cisco huawei, dan melakukan trace lagi untuk memastikan jalur routing kembali pindah ke link utama cisco-huawei.

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.377 ms  4.617 ms  5.323 ms
 2  10.10.10.2     5.596 ms  4.418 ms  5.219 ms
 3  *192.168.50.10  9.551 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 44: trace 192.168.50.10

2 Analisis Hasil Percobaan

Berdasarkan hasil praktikum yang telah dilakukan, seluruh VLAN yang dikonfigurasi pada Cisco Switch berhasil dibuat dan berada dalam kondisi aktif. Hal ini ditunjukkan oleh hasil perintah *show vlan brief* yang menampilkan VLAN 10, VLAN 20, VLAN 30, dan VLAN 40. Konfigurasi access port dan trunk port juga berhasil dilakukan sehingga komunikasi antara switch dan router dapat berjalan sesuai konsep *router-on-a-stick*.

Pada tahap konfigurasi Cisco Router, DHCP Server berhasil memberikan alamat IP secara otomatis kepada host pada VLAN 10 dan VLAN 20. Keberhasilan tersebut dapat dilihat dari hasil *show ip dhcp binding* dan pengujian pada VPCS yang memperoleh alamat IP sesuai dengan jaringan masing-masing. Selain itu, seluruh host dapat melakukan ping ke gateway sehingga menunjukkan bahwa konfigurasi subinterface dan gateway telah berjalan dengan baik.

Pengujian komunikasi antar-VLAN menunjukkan hasil yang berhasil. Host pada VLAN 10 dapat berkomunikasi dengan host pada VLAN 20, VLAN 30, dan VLAN 40. Hasil ini sesuai dengan teori inter-VLAN routing yang memanfaatkan router Layer 3 untuk meneruskan paket antar jaringan yang berbeda.

Pada konfigurasi jaringan multivendor, Cisco Router, Huawei Router, dan MikroTik berhasil saling terhubung melalui link antar-router yang telah dikonfigurasi. Hasil pengujian ping antar-router menunjukkan bahwa seluruh interface aktif dan dapat saling berkomunikasi. Hal ini menunjukkan bahwa konfigurasi IP address dan subnet mask telah sesuai.

Implementasi OSPF pada perangkat Cisco, Huawei, dan MikroTik berhasil membentuk hubungan *neighbor adjacency*. Masing-masing router dapat mempelajari jaringan yang berasal dari router lain melalui pertukaran informasi routing OSPF. Selain itu, routing table menunjukkan bahwa jalur dengan nilai cost terendah dipilih sebagai jalur utama sesuai dengan teori OSPF.

Pada pengujian failover, ketika salah satu link utama Cisco-Huawei dimatikan, jalur komunikasi secara otomatis berpindah ke link alternatif yang masih tersedia. Ketika kedua link utama dimatikan, trafik dialihkan melalui MikroTik sebagai jalur cadangan. Setelah kedua link diaktifkan kembali, OSPF kembali memilih jalur utama yang memiliki cost lebih rendah. Hasil ini menunjukkan bahwa proses konvergensi OSPF berjalan dengan baik dan mampu menjaga ketersediaan jaringan saat terjadi kegagalan link.

Ringkasan hasil pengujian dapat dilihat pada Tabel 1. Secara keseluruhan, seluruh tujuan praktikum berhasil dicapai dan hasil yang diperoleh telah sesuai dengan teori mengenai VLAN, DHCP, inter-VLAN routing, OSPF multivendor, dan failover routing.

Tabel 1: Ringkasan Hasil Pengujian

Pengujian	Hasil	Keterangan
DHCP VLAN 10	Berhasil	IP diperoleh otomatis dari DHCP Server
DHCP VLAN 20	Berhasil	IP diperoleh otomatis dari DHCP Server
Ping ke Gateway	Berhasil	Tidak ada packet loss
Ping Antar VLAN	Berhasil	Inter-VLAN Routing berjalan dengan baik
OSPF Neighbor	Berhasil	Adjacency terbentuk antar-router
Ping Cisco-Huawei	Berhasil	Routing OSPF berjalan dengan baik
Failover Link 1 Putus	Berhasil	Trafik berpindah ke jalur alternatif
Failover Link 1 dan 2 Putus	Berhasil	Trafik dialihkan melalui MikroTik
Recovery Link	Berhasil	Jalur kembali ke link utama

3 Hasil Tugas Modul

1. Tugas modul 5 (kelompok) sudah diunggah pada Github bagian LA/README.md

4 Kesimpulan

Praktikum ini berhasil mengimplementasikan VLAN, inter-VLAN routing, DHCP Server, serta routing dinamis OSPF pada lingkungan multivendor yang terdiri dari Cisco, Huawei, dan MikroTik. Seluruh host dapat memperoleh alamat IP, berkomunikasi dengan gateway, melakukan komunikasi antar-VLAN, serta mengakses jaringan pada router lain melalui OSPF.

Hasil pengujian menunjukkan bahwa OSPF mampu membentuk *neighbor adjacency*, mendistribusikan informasi routing, dan melakukan failover secara otomatis ketika terjadi kegagalan link. Dengan demikian, seluruh tujuan praktikum telah tercapai dan hasil yang diperoleh sesuai dengan teori yang dipelajari mengenai segmentasi jaringan, routing dinamis, dan redundansi jaringan.

5 Lampiran

5.1 Dokumentasi saat praktikum

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.377 ms  4.617 ms  5.323 ms
 2  10.10.10.2     5.596 ms  4.418 ms  5.219 ms
 3  *192.168.50.10  9.551 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 45: Salah satu dokumentasi hasil pengujian failover OSPF ketika dua link diaktifkan kembali